Troy Mayer

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EDUCATION

Purdue University, School of Mechanical Engineering and School of Science **West Lafayette, IN**
B.S. Mechanical Engineering; B.S. Applied Statistics; Minor in Economics; Degree GPA 3.4/4.0 **May 2028**

PROFESSIONAL EXPERIENCE

Boston Dynamics **Waltham, MA**
Sensors and Perception Co-Op – Atlas | *Python, MuJoCo, Flask, SQLite, CATIA, Linux* **July 2026 – Present**

- Built a 3D impact-energy model using Khatib's operational-space formulation, estimating each fall strike's energy from logged joint states, the mass matrix and reflected rotor inertia, verified against hand-derived Jacobians.
- Designing CATIA fixtures, led by drop rigs that validate it: a rail-guided die spring on a two-axis gimbal centered on the contact point, aimed along the strike's momentum so nothing slides off and it absorbs the full 3D energy.
- Built an impact-forensics ETL pipeline and Flask/Plotly dashboard that loads ~900 robot test logs into SQLite, giving engineers a 3D heatmap of ~28k impacts on the robot's mesh and per-event waveforms during live testing.
- Modeled hardware thermal behavior in Python, fitting heating curves with Savitzky-Golay filtering and multi-start least squares to predict steady-state temperatures across hundreds of units without full-length tests.

Chortos Lab – Soft Adaptive Systems and Structures **West Lafayette, IN**
Undergraduate Researcher | *Python, LabVIEW, MATLAB, Fusion 360* **August 2025 – May 2026**

- Designed an experiment to characterize the stress–strain response of ionic dielectric gels under voltage, building 15 Fusion 360 fixtures to try new sample geometries under identical conditions.
- Automated LabVIEW and MATLAB acquisition from a Keyence laser displacement head via an NI USB-6001, and wrote a Python computer-vision tracker measuring cantilever deflection to 10 µm against an in-frame micrometer scale.
- Formulated, synthesized and direct-ink-write printed gel variants, using those tests to tune their chemistry until electromechanical force output rose by over 50%.

Emerson – ASCO **Florham Park, NJ**
Mechanical and Manufacturing Engineering Intern | *Creo, CNC, GD&T* **May 2025 – August 2025**

- Designed 30+ rotary valve components in Creo, verified by hand stress analysis, and machined pistons to ±0.05 mm.
- Root-caused actuator deadband from cycle-test data and wrote an assembly procedure that cut it by over 10%.

PROJECTS

Purdue ASME – Biomechanics **West Lafayette, IN**
Active Orthosis – Electronics Team Lead | *C++17, FreeRTOS, KiCad, GoogleTest* **April 2024 – May 2026**

- Designed a 1 kHz feed-forward PI velocity controller with anti-windup in C++17 on an ESP32-S3, tuned from a first-order motor model to hold simulated speed error to 0.8%, guarded by 10 latching fault monitors and a watchdog.
- Cut EMG-to-motion latency from 103 to 43 ms with a Bayesian estimator, simulated on 24 real recorded contractions.
- Led a 5-engineer team redesigning the 4-channel EMG board in KiCad with an INA333 instrumentation front end, right-leg drive and ISO7763 motor isolation, raising day-to-day gesture decoding from 31% to 87% in emulated tests.

Asteroid Hopping Robot – Independent Research
Purdue University & Universidad Carlos III de Madrid | *Fusion 360, 3D Printing* **October 2026 – Present**

- Designed the full robot as a parametric Fusion 360 assembly: three five-bar legs driven by GM2804 gimbal motors through 3:1 timing belts, two internal reaction wheels, and an XL330 servo arm with a differential wrist and gripper.
- Building it to reshape its legs between hops, with push-off control optimized for a target launch speed, leg impedance control to absorb landings and reaction-wheel control to reorient mid-flight.
- Developing a pendulum landing rig and risk-limited hop policies that trade distance against escape and tip-over.

Legged Robot State Estimation and Control
Personal Project | *C++, Eigen, MuJoCo, GoogleTest, Python* **August 2026 – October 2026**

- Built a contact-aided error-state Kalman filter in C++/Eigen that fuses a simulated quadruped's 1 kHz IMU with leg kinematics, estimating base velocity to 0.013 m/s RMSE against 1.38 m/s for IMU-only.
- Verified every Jacobian against finite differences, then ran a Raibert trot controller and a QP balance controller on the estimate instead of ground truth, walking 5 seeds with zero falls and velocity tracking within 0.004 m/s.

Caterpillar (via Purdue Data Mine) **West Lafayette, IN**
Machine Learning Researcher | *Python, Snowflake, Power BI* **January 2026 – May 2026**

- Consolidated five years of noisy paint-line data and built LSTM and XGBoost forecasts of primer, topcoat and solvent use, contributing to a 4.7% cut in material usage and a 3.3% cut in cycle time.

SKILLS & INTERESTS

- **Tools:** C++, Python, MATLAB, SQL, LabVIEW, Eigen, MuJoCo, FreeRTOS, GoogleTest, KiCad, CATIA, Creo, Linux
- **Languages:** English (Native), Turkish (Fluent), Spanish (Fluent - New Jersey Seal of Biliteracy)
- **Clubs & Hobbies:** Club Swimming, House Band, Lifting, Hiking, Guitar, Saxophone